

Assignment 5 of MATP4820

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March 19, 2026

Problem 1

Find the projection of a given point $\mathbf{x} \in \mathbb{R}^n$ onto the set X . Justify your solution. For Questions 1 and 2, sketch the set X and plot where the projected point is on X .

1. $X = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, 2x_1 + x_2 \leq 2\}; \mathbf{x} = [1; 2]$

First, $x_1 = 1$ and $x_2 = 2 \implies x_1 \geq 0, x_2 \geq 0, 2\mathbf{x}_1 + \mathbf{x}_2 > 2 \implies x \notin X$

Thus, the active constraint is $2x_1 + x_2 \leq 2$.

Now,

$$\text{Proj}_X(x) = \arg \min_{y \in X} \|y - x\|_2 = \min_{y_1, y_2} \sqrt{(y_1 - 1)^2 + (y_2 - 2)^2}$$

$$\text{Subject to } 2y_1 + y_2 = 2 \implies y_2 = 2 - 2y_1$$

Substituting $y_2 = 2 - 2y_1$

$$\min_{y_1} \sqrt{(y_1 - 1)^2 + (2 - 2y_1 - 2)^2} = \sqrt{y_1^2 - 2y_1 + 1 + 4y_1^2} = \sqrt{5y_1^2 - 2y_1 + 1}$$

Minimizing $\sqrt{5y_1^2 - 2y_1 + 1}$ is the same as minimizing $f(y_1) = 5y_1^2 - 2y_1 + 1$.

We minimize $f(y_1)$ by setting $f'(y_1) = 0$

$$f(y_1) = 5y_1^2 - 2y_1 + 1$$

$$\implies f'(y_1) = 10y_1 - 2$$

$$f'(y_1) = 0 \implies 10y_1 - 2 = 0 \implies y_1 = 0.2$$

Now,

$$y_2 = 2 - 2(0.2) = 1.6$$

Thus we get $\text{Proj}_X(x) = \begin{bmatrix} 0.2 \\ 1.6 \end{bmatrix} = [0.2; 1.6]$

Sketch:

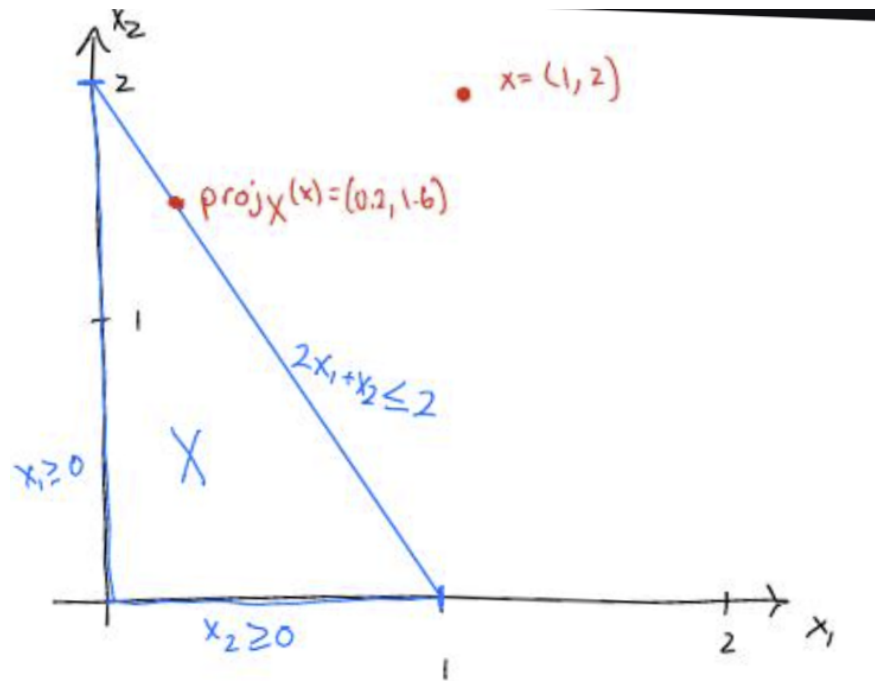


Figure 1: Sketch of the set X and the projected point $\text{Proj}_X(x)$ for Question 1

$$2. X = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \leq 0, x_1^2 + x_2^2 \leq 1\}; \mathbf{x} = [-2; 2]$$

$$\text{First, } x_1 = -2 \text{ and } x_2 = 2 \implies x_1 \leq 0, x_1^2 + x_2^2 \geq 1, (-2)^2 + 2^2 > 1 \implies x \notin X$$

Now as derived in lecture,

$$X = \{x \in \mathbb{R}^2 : |x|_2 \leq \lambda,$$

and

$$\text{Proj}_X(x) = \begin{cases} x & \text{if } |x|_2 \leq \lambda \\ \frac{\lambda x}{|x|_2} & \text{if } |x|_2 > \lambda \end{cases}$$

Where

$$x = [-2; 2], \lambda = 1 \implies |x|_2 = \sqrt{(-2)^2 + (2)^2} = \sqrt{8} = 2\sqrt{2}$$

Now $|x|_2 = 2\sqrt{2} > 1$, so

$$\text{Proj}_X(x) = \frac{1}{2\sqrt{2}} \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \frac{x}{2\sqrt{2}} = \frac{1}{2\sqrt{2}} \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Sketch:

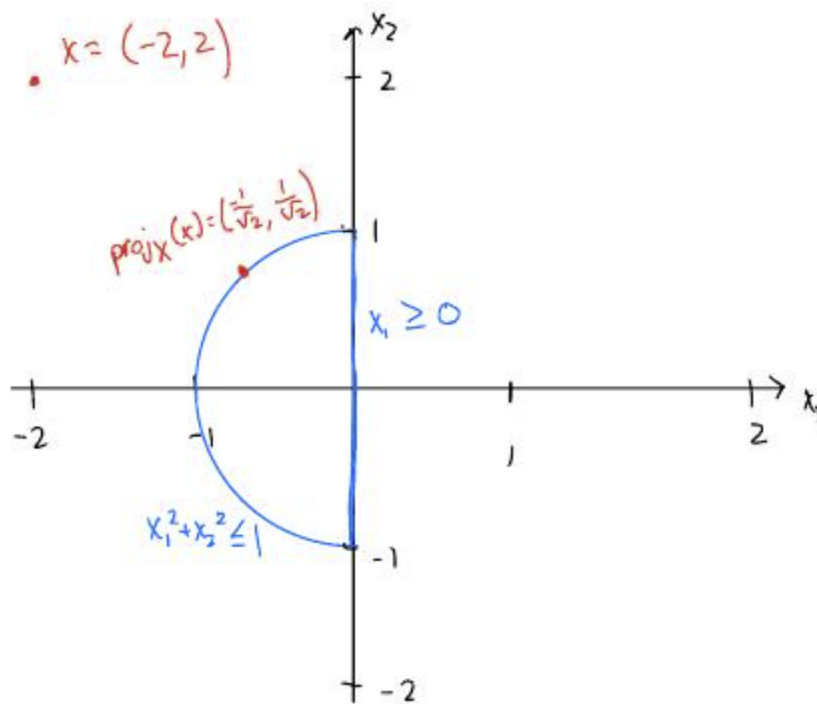


Figure 2: Sketch of the set X and the projected point $\text{Proj}_X(x)$ for Question 2

Problem 2

Let $g(\mathbf{x}) = 2\|\mathbf{x}\|_2$ for $\mathbf{x} \in \mathbb{R}^n$.

1. Let $n = 2$. Evaluate the proximal mapping $\mathbf{prox}_g$ at $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$$\text{Prox}_g(x) = \arg \min_{y \in \mathbb{R}^n} \left(\frac{1}{2} \|y - x\|_2^2 + g(y) \right)$$

Take $n = 2$, $g(y) = 2\|y\|_2$, $x = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Then,

$$\text{Prox}_g(x) = \arg \min_{y \in \mathbb{R}^n} \left(\frac{1}{2} \left\| y - \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\|_2^2 + 2\|y\|_2 \right)$$

Now take

$$f(x) = \frac{1}{2} \left\| y - \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\|_2^2 + 2\|y\|_2$$

We minimize $f(x)$ by setting $f'(x) = 0$

$$\begin{aligned} f'(x) &= y - \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 2 \frac{y}{\|y\|_2} \\ \implies y - \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 2 \frac{y}{\|y\|_2} &= 0 \\ y \left(1 + \frac{2}{\|y\|_2} \right) &= \begin{bmatrix} -1 \\ 2 \end{bmatrix} \end{aligned}$$

Taking norm of both sides

$$\left\| y \left(1 + \frac{2}{\|y\|_2} \right) \right\|_2 = \left\| \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\|_2$$

Pulling out $1 + \frac{2}{\|y\|_2}$ (Scalar)

$$\begin{aligned} \implies \left(1 + \frac{2}{\|y\|_2} \right) \|y\|_2 &= \sqrt{5} \\ \|y\|_2 + 2 &= \sqrt{5} \\ \|y\|_2 &= \sqrt{5} - 2 \end{aligned}$$

Now, as we previously derived,

$$y \left(1 + \frac{2}{\|y\|_2} \right) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Subbing in $\sqrt{5} - 2$

$$\implies y \left(1 + \frac{2}{\sqrt{5} - 2} \right) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\implies y = \left(1 + \frac{2}{\sqrt{5} - 2} \right)^{-1} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \left(\frac{\sqrt{5} - 2 + 2}{\sqrt{5} - 2} \right)^{-1} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \left(\frac{\sqrt{5}}{\sqrt{5} - 2} \right)^{-1} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \left(1 - \frac{2}{\sqrt{5}} \right) \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Thus,

$$y = \left(1 - \frac{2}{\sqrt{5}} \right) \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 + \frac{2}{\sqrt{5}} \\ 2 - \frac{4}{\sqrt{5}} \end{bmatrix}$$

Problem 3

Consider the unconstrained quadratic minimization problem:

$$\underset{\mathbf{x}}{\text{minimize}} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad (\text{QuadMin})$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a symmetric positive definite matrix, and $\mathbf{b} \in \mathbb{R}^n$.

Write a solver for (QuadMin) by the DFP method. Use the instructor's provided file `quadMin_DFP.m` to write a Matlab function `quadMin_DFP` with input $\mathbf{A}$, $\mathbf{b}$, initial vector $\mathbf{x}_0$, and tolerance `tol`, and with stopping condition $\|\nabla f(\mathbf{x}^k)\| \leq \text{tol}$. Also test your function by running the provided test file `test_DFP.m` and compare to the instructor's function. Print your code and the results you get, and print the figures you get.

DFP method Function

```
function [x, hist_res] = quadMin_DFP(A,b,x0,tol)

% DFP method for solving
% min_x 0.5*x'*A*x - b'*x

x = x0;

%% perform one steepest gradient descent with exact line search

% compute gradient of the objective
grad = A * x - b ;

% evaluate the norm of gradient
res = norm(grad);

% save the value of res
hist_res = res;

% choose the initial H matrix

H = eye(length(b));

%% main iteration
```

```

while res > tol

    % compute the search direction
    p = -H * grad;

    % use exact line search
    alpha = norm(p)^2 / ( p'*(A*p) );

    % update x
    x0 = x
    x = x + alpha * p;

    grad0 = grad;
    % compute gradient of the objective
    grad = A * x - b;

    % compute s and y
    s = x - x0;

    y = grad - grad0;

    % update H matrix
    H = H - (H*y*y'*H)/(y'*H*y) + (s*s')/(s'*y);

    % evaluate the norm of gradient
    res = norm(grad);

    % save the value of res
    hist_res = [hist_res; res];
end

end

```

Code Results: **test_DFP.m** Output

```
>> test_DFP
Student solver: Total running time is 34.4173
Final objective value is -5.8974
Instructor solver: Total running time is 7.7950
Final objective value is -5.8974
```

Student Figure Result

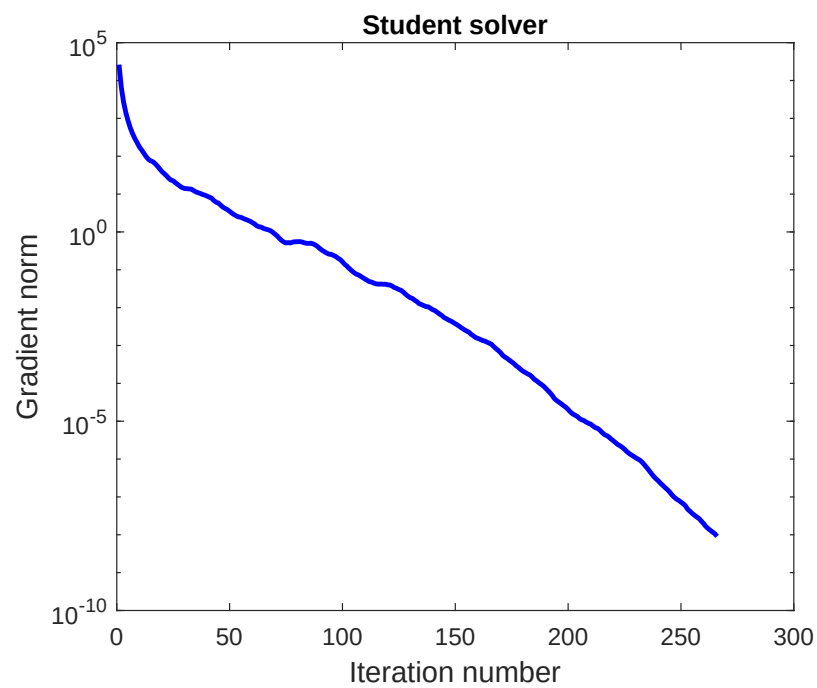


Figure 3: Student DFP Result

Instructor Figure Result

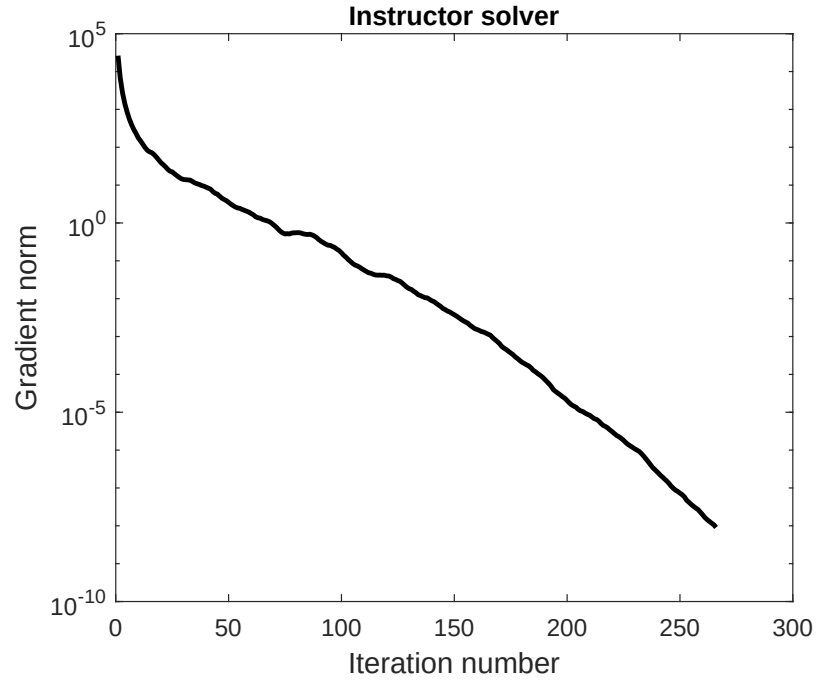


Figure 4: Instructor DFP Result

Our proposed DFP method reaches the same final objective value in the same number of iterations shown in the figure. While the instructor's implementation converges noticeably faster, this difference is likely due to more optimized operations, function choices, and memory management rather than a difference in implementation logic.